

The Effect of Expected Income on Individual Migration Decisions

John Kennan and James R. Walker

Tate Mason

John Munro Godfrey Sr. Dept. of Economics
University of Georgia

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Roadmap (Talk-First Design)

- ① Motivation & Stylized Facts
- ② Model Intuition
- ③ Estimation Idea
- ④ Results: What Drives Migration?
- ⑤ Interpretation & Policy Angles
- ⑥ Q&A (Appendix: all proofs/derivations/robustness)

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Question & Motivation

- **Question:** Do expected income differences drive interstate migration?
- Migration is common for the young; repeat and return moves are pervasive.
- Policy relevance: local shocks, subsidies, welfare differentials.

Stylized Facts (NLSY79)

TABLE I
INTERSTATE MIGRATION, NLSY 1979–1994^a

	Less Than High School	High School	Some College	College	Total
Number of people	322	919	758	685	2,684
Movers (age 18–27)	80	223	224	341	868
Movers (%)	24.8%	24.3%	29.6%	49.8%	32.3%
Moves per mover	2.10	1.95	1.90	2.02	1.98
Repeat moves (% of all moves)	52.4%	48.7%	47.4%	50.5%	49.5%
Return migration (% of all moves)					
Home	32.7%	33.1%	29.1%	23.2%	28.1%
Not home	15.5%	7.1%	6.8%	8.6%	8.4%
Movers who return home	61.3%	56.5%	51.3%	42.8%	50.2%

^aThe sample includes respondents from the cross-section sample of the NLSY79 who were continuously interviewed from ages 18 to 28 and who never served in the military. The home location is the state of residence at age 14.

- Repeat moves are about half of all moves; many are returns to origin.
- Movers are younger; return migration is substantial.

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Model Basics

- **Setup:** Locations differ in mean wages and match quality; moving is costly.
- **Information:** Current location wage known; other locations' matches known only by visiting (experience good).
- **Decision:** Maximize expected lifetime income net of moving costs.

State at t :

current location ℓ_0 , known w_{ℓ_0}

prob. $\mathbb{E}(v(\text{stay}))$

prob. $\mathbb{E}(v(j))$

Stay

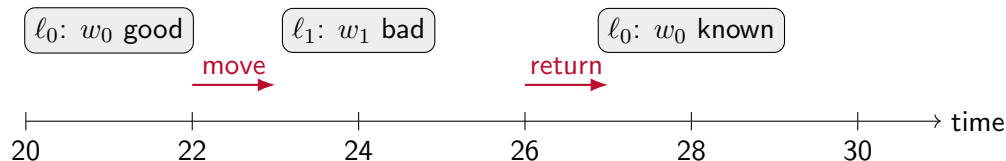
Move to $j \neq \ell_0$

$$u_t = \alpha_0 w_{\ell_0} + \text{amenities} + \zeta_{\ell_0} \\ + \beta \mathbb{E}[V_{t+1} | \text{age} + 1, \ell_0]$$

$$u_t = \alpha_0 w_j - \Delta(\ell_0, j) \\ + \text{amenities} + \zeta_j \\ + \beta \mathbb{E}[V_{t+1} | \text{age} + 1, j]$$

Why Dynamic & Many Locations?

- Return migration and onward moves require dynamics.
- Many alternatives matter: returning is a choice among many states.



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Computational Challenge → Limited-History Approximation

- Full state space explodes with locations and match histories.
- **Idea:** Keep only the most recent M locations for wages/matches.
- Preserves behavior seen in the data while making DP feasible.

Full state (infeasible):
(all past locations, $\{\text{all } w, \xi\}$)

truncate to last M

Limited history (M recent):
 $((\ell_0, \ell_1), (w_{\ell_0}, w_{\ell_1}), (\xi_{\ell_0}, \xi_{\ell_1}))$

Wage Components (What Matters for Moves)

- Mean wage by location; permanent match component; person FE; transitory shocks.
- Migration choice depends on *differences* in expected income and moving cost, not levels.

Wage decomposition (relevant pieces in bold)

$$\underbrace{w_{ij}(a, t)}_{\text{observed}} = \underbrace{\mu_j}_{\text{mean by location}} + \underbrace{v_{ij}}_{\text{match}} + G(X_i, a, t) + \eta_i + \varepsilon_{ijt}$$

- Migration choice depends on $\mu_j + v_{ij}$ and moving cost $\Delta(\ell_0, j)$.

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What Drives Migration? (Headlines)

- **Expected income matters:** moves respond to geographic mean wages and bad current matches.
- **Costs matter:** distance penalty; strong home premium; cheaper to return.
- **Population size** of destination reduces cost (network/proximity channels).

TABLE II
INTERSTATE MIGRATION, YOUNG WHITE MEN^a

	$\hat{\theta}$	$\hat{\sigma}_{\theta}$	$\hat{\theta}$	$\hat{\sigma}_{\theta}$	$\hat{\theta}$	$\hat{\sigma}_{\theta}$	$\hat{\theta}$	$\hat{\sigma}_{\theta}$
<i>Utility and cost</i>								
Disutility of moving (γ_0)	4.790	0.565	4.514	0.523	4.864	0.601	4.851	0.604
Distance (γ_1) (1000 miles)	0.265	0.182	0.280	0.178	0.311	0.187	0.270	0.184
Adjacent location (γ_2)	0.808	0.214	0.787	0.211	0.773	0.220	0.804	0.216
Home premium (α^H)	0.331	0.041	0.267	0.031	0.332	0.047	0.337	0.045
Previous location (γ_3)	2.757	0.356	2.544	0.300	3.082	0.449	2.818	0.416
Age (γ_4)	0.055	0.020	0.062	0.019	0.060	0.020	0.054	0.020
Population (γ_5) (millions)	0.653	0.179	0.653	0.178	0.635	0.177	0.652	0.179
Stayer probability	0.510	0.078	0.520	0.079	0.495	0.087	0.508	0.082
Cooling (α_1) (1000 degree-days)	0.055	0.019	0.036	0.019	0.048	0.018	0.056	0.019
Income (α_0)	0.312	0.100	—	—	—	—	0.297	0.116
Location match preference (τ_{ξ})	—	—	—	—	0.168	0.049	0.070	0.099
<i>Wages</i>								
Wage intercept	-5.165	0.244	-5.175	0.246	-5.175	0.246	-5.168	0.244
Time trend	-0.035	0.008	-0.033	0.008	-0.033	0.008	-0.035	0.008
Age effect (linear)	7.865	0.354	7.876	0.356	7.877	0.356	7.870	0.355
Age effect (quadratic)	-2.364	0.129	-2.381	0.130	-2.381	0.130	-2.367	0.129
Ability (AFQT)	0.014	0.065	0.015	0.066	0.014	0.066	0.014	0.065
Interaction (Age, AFQT)	0.147	0.040	0.152	0.040	0.152	0.040	0.147	0.040
Transient s.d. 1	0.217	0.007	0.218	0.007	0.218	0.007	0.217	0.007
Transient s.d. 2	0.375	0.015	0.375	0.015	0.375	0.015	0.375	0.015
Transient s.d. 3	0.546	0.017	0.547	0.017	0.547	0.017	0.546	0.017
Transient s.d. 4	1.306	0.028	1.307	0.028	1.307	0.028	1.306	0.028
Fixed effect 1	0.113	0.035	0.112	0.035	0.112	0.035	0.113	0.035
Fixed effect 2	0.298	0.035	0.296	0.035	0.296	0.035	0.298	0.035
Fixed effect 3	0.936	0.017	0.934	0.017	0.934	0.017	0.936	0.017
Wage match (τ_v)	0.384	0.017	0.387	0.018	0.387	0.018	0.384	0.018
<i>Log likelihood</i>								
Exclude income: $\chi^2(1)$	-4,214.880		-4,221.426		-4,218.800		-4,214.834	
			13.09		7.93			
Exclude match preference: $\chi^2(1)$		0.09		5.25				

^aThere are 4,274 (person-year) observations, 432 individuals, and 124 moves.

Economic Magnitudes

- Moving cost equivalent $\approx$ \$300k (PV terms, order of magnitude).
- Good match + high-mean state $\Rightarrow \approx$ \$22k / year gain (1989).

Tables of Magnitudes

TABLE III
WAGE PARAMETER ESTIMATES (IN 2010 DOLLARS)

	AFQT Percentile		
	25	50	75
<i>Average wages</i>			
Age 20 in 1979	25,827	27,522	29,216
Age 20 in 1989	18,472	20,166	21,861
Age 30 in 1989	40,360	42,850	45,340
	Low	Middle	High
Location match	−8,366	0	8,366
Fixed effect support	−20,411	−6,498 −2,454	0 2,454 6,498
State means	Low (WV) 12,698	Rank 5 (OK) 14,530	Median (MO) 16,978
			Rank 45 (RI) 19,276
			High (MD) 22,229

TABLE IV
MOVING COST EXAMPLES

	γ_0	α_0	Age	Distance	Adjacent	Population	Previous	Cost
θ	4.790	0.312	0.055	0.265	0.808	0.653	2.757	
Young mover			20	1	0	1	0	\$384,743
Average mover			24.355	0.664	0.427	0.728	0.371	\$312,146

Fit & Mechanism

- Model reproduces repeat/return migration shares qualitatively.
- Negative association: higher realized income this period $\Rightarrow$ lower out-migration probability.

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What We Learn

- Migration resembles optimal search over locations.
- Both mean differences and idiosyncratic match quality drive moves.
- Structural approach enables policy counterfactuals.

Policy Angles & Extensions

- Local labor demand shocks & wage equalization dynamics.
- Subsidies or relocation assistance: who moves more?
- Household joint decisions; welfare benefit differentials; retirement migration.

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Takeaways

- Expected income plays a sizable role in interstate migration.
- High moving costs + payoff shocks reconcile low average move rates with meaningful gains.
- Model provides a tractable template for counterfactual policy analysis.

Thanks!

Questions? tate.mason@uga.edu

Roadmap

7 Appendix

Data & Sample Construction (*Appendix*)

- NLSY79 cross-section; white non-Hispanic HS grads; ages 20+; not in military.
- Attrition and consistency checks; AFQT use; state means from Census PUMS (1990).

Sample Selection

	Respondents	Person-Years
White non-Hispanic males (cross-section sample)	2,439	39,024
<i>Restrictions applied to respondents</i>		
Ever in military	-246	
High school dropouts and college graduates	-1,290	
Attended college	-130	
Older than age 20 at start of sample period	-134	
Missing AFQT score	-41	
Attend or graduate from high school at age 20	-87	
Not in labor force for more than 1 year after age 19	-44	
Location at age 20 not reported	-20	
Income information inconsistent	-1	
Died before age 30	-4	
Residence at age 14 not reported	-2	
In jail in 1993	-1	
Subtotal	-2,000	6,585
<i>Restrictions applied to periods</i>		
Delete periods after first gap in history	-1	-1,104
Delete periods before age 20	-6	-1,207
Analysis sample	432	4,274

Sample Selection — Years per person

	Respondents	Person-Years
<i>Years per person</i>		
1	14	14
2	16	32
3	19	57
4	14	56
5	14	70
6	14	84
7	13	91
8	9	72
9	34	306
10	61	610
11	53	583
12	44	528
13	45	585
14	44	616
15	38	570
Total	432	4,274

Value Function & Choice Probabilities (*Appendix*)

Bellman equation

$$V(x, \zeta) = \max_{j \in \mathcal{J}} \{ v(x, j) + \zeta_j \},$$
$$v(x, j) = u(x, j) + \beta \mathbb{E}_{x'} [\bar{V}(x') \mid x, j], \quad \bar{V}(x) = \mathbb{E}_{\zeta} [V(x, \zeta)].$$

Extreme value shocks $\Rightarrow$ logit choice probabilities

$$\rho(x, j) = \frac{\exp(v(x, j))}{\sum_{k \in \mathcal{J}} \exp(v(x, k))}, \quad \exp(\bar{V}(x)) = e^{\bar{\gamma}} \sum_k \exp(v(x, k)).$$

Limited-History State Transitions (*Appendix*)

State and transitions (with $M = 2$ recent locations)

State: $x = (\ell_0, \ell_1, \omega_{\ell_0}^{(w)}, \omega_{\ell_1}^{(w)}, \omega_{\ell_0}^{(\xi)}, \omega_{\ell_1}^{(\xi)}, a)$.

If stay: $x' = (\ell_0, \ell_1, \cdot, \cdot, \cdot, \cdot, a + 1)$.

If move to previous ℓ_1 : swap (ℓ_0, ℓ_1) and their components, $a \leftarrow a + 1$.

If move to new j : shift $(\ell_0 \leftarrow j, \ell_1 \leftarrow \ell_0)$ and draw new $(\omega_j^{(w)}, \omega_j^{(\xi)})$.

Likelihood & Identification Sketch (*Appendix*)

- Joint likelihood of income & choices; mixture over unobserved components.
- Nonparametric intuition: selection on bad current draws $\uparrow$ move probability.

Robustness (*Appendix*)

- Vary β , horizon T , include preference match, extra amenities.
- Fit diagnostics; simulated vs. observed migration rates.

Robustness (Appendix)

TABLE XIV
ESTIMATES FROM SIMULATED MIGRATION HISTORIES

	Base Model		100 Reps		<i>t</i>
	$\hat{\theta}$	$\hat{\sigma}_{\theta}$	$\hat{\theta}$	$\hat{\sigma}_{\theta}$	
Disutility of moving	4.790	0.565	4.732	0.058	-1.00
Distance	0.265	0.182	0.272	0.015	0.42
Adjacent location	0.808	0.214	0.808	0.017	0.04
Home premium	0.331	0.041	0.333	0.004	0.42
Previous location	2.757	0.356	2.709	0.031	-1.52
Age	0.055	0.020	0.056	0.002	0.25
Population	0.653	0.179	0.648	0.017	-0.30
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Income	0.312	0.100	0.307	0.008	-0.56
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Time trend	-0.035	0.008	-0.035	0.001	0.40
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Age effect (quadratic)	-2.364	0.129	-2.359	0.018	0.28
Ability (AFQT)	0.014	0.065	0.017	0.010	0.35
Interaction (age, AFQT)	0.147	0.040	0.142	0.007	-0.67
Transient s.d. 1	0.217	0.007	0.217	0.001	0.07
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Transient s.d. 3	0.546	0.017	0.545	0.002	-0.84
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Fixed effect 2	0.298	0.035	0.300	0.003	0.68
Fixed effect 3	0.936	0.017	0.938	0.002	1.37
Wage match	0.384	0.017	0.383	0.002	-0.17
Log likelihood, $\chi^2(24)$	-4,214.88		-473,830.3		9.95